



Basic Emergency Care

Fractures, sprains, and strains

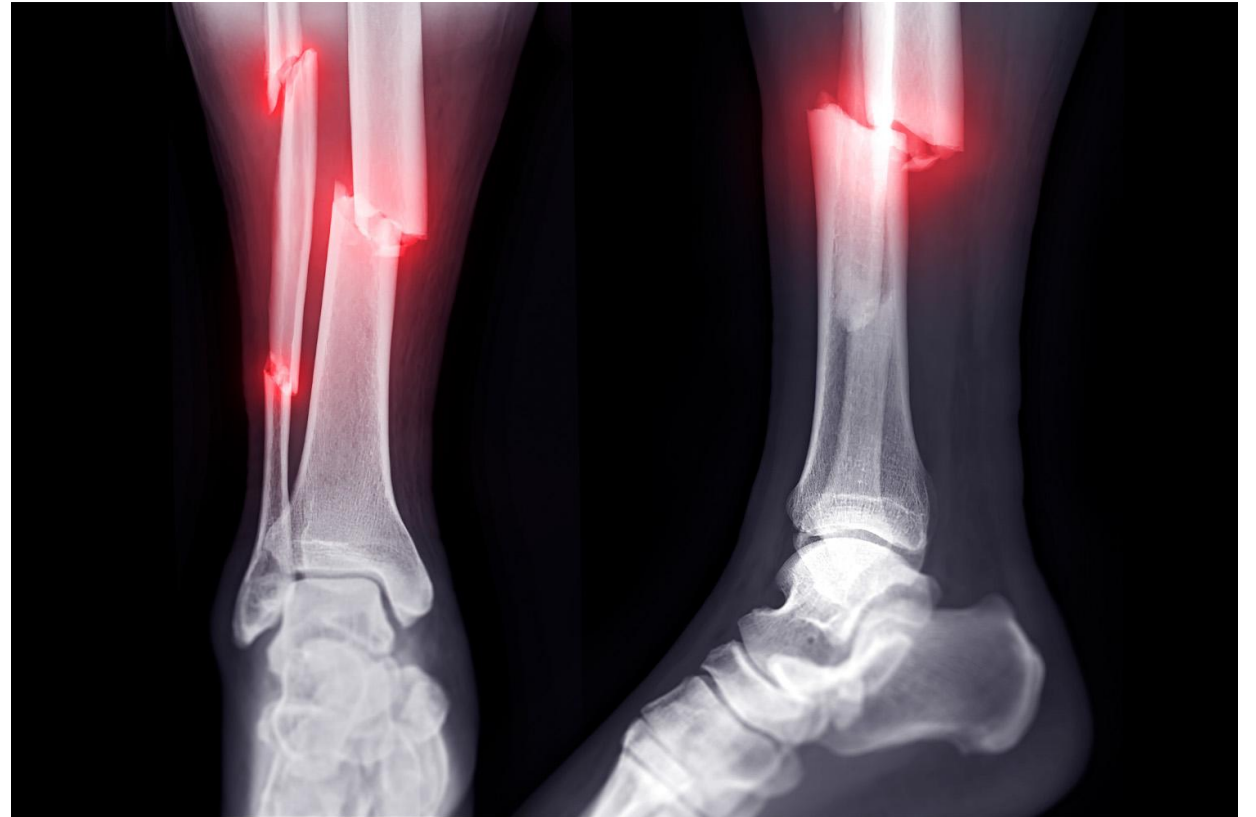
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Learning objectives

- Define fracture, sprain, and strain
- Differentiate between them clinically
- Perform initial emergency assessment (ABCDE)
- Identify red flags
- Provide basic first aid and immobilization

Fracture

- **Fracture:**
- Is a break in the cortical continuity of a bone.



Fracture

- **Causes:**
- Road traffic accidents
- Falls
- Sports injuries
- Direct trauma
- Twisting injury
- Osteoporosis (elderly)

Fracture

- Classification:

Closed (Simple)

Skin intact.

Open (Compound)

Bone communicates with outside

→ infection risk

Displaced

Bone ends are aligned

Non displaced

Bone end are not aligned

Complete

Break of continuity on both sides

Incomplete (greenstick)

Break of continuity on one side

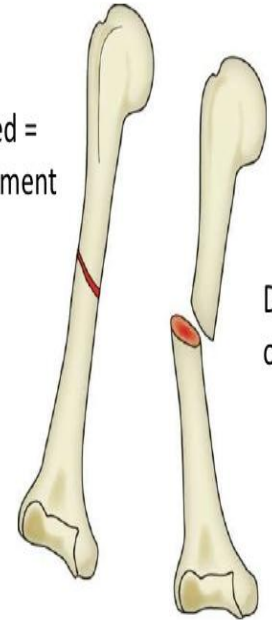
Fracture



b.

Displaced and Non displaced

Non displaced =
normal alignment



Displaced = pulled out
of normal alignment

Fracture

- **Presentation:**

- Severe pain
- Swelling
- Bruising
- Deformity
- Shortening of limb
- Abnormal movement
- Inability to bear weight

Fracture

- Management:
- D: Danger
- R: Response
- A: Airway
- B: Breathing
- C: Circulation
- D: Disability
- E: Exposure

Fracture

- **Danger:** Make sure it is safe to reach the patient
- **Response:** Quick checking of patient's status
- **Airway:** Ensure patent airway (Left lateral position, remove any foreign body in the mouth, basic airway maneuvers) + C-spine
- **Breathing:** Check respiratory status and any chest injury, Give O2
- **Circulation:** Check BP, HR, CRT, 2 large IV cannula, control any external bleeding
- **Disability:** Check GCS, RBS, Neurological status. Decreased LOC could have been a cause for the fracture.
- **Exposure:** Head to toe examination for any additional injury

Fracture

- **Neurovascular assessment:**
- Comes after initial resuscitation.
- A must!!!
- To detect or exclude any life threatening or limb threatening conditions
- Check: Pulse, capillary refill time, sensation, and movement distal to the pulse

Fracture

Management of fracture:

- Immobilize the limb
- Use splints
- Elevate limb
- Give analgesia
- Do not attempt aggressive realignment
- Do not push exposed bone inside

Cervical fracture

- **They are Dangerous:**

- Spinal cord injury
- Paralysis
- Respiratory failure
- Death

- **Mechanism of injury:**

Intense high velocity traumas such as:

- High speed Collision
- Fall from great height
- Diving

Cervical fracture

- **Red flags for cervical fracture in trauma**

- ⚠ Neck pain after trauma
- ⚠ Loss of consciousness
- ⚠ Neurological deficit
- ⚠ High-speed accident
- ⚠ Tingling or weakness in limbs



Cervical fracture

- **Management of cervical fracture:**
- Starts during the A!
- Immobilize the neck with rigid cervical collar
- Follow the rest of ABCDE
- Refer urgently



Pelvic fracture

- **They are Dangerous:**

- Major blood vessels
- Internal organs (bladder, bowel)

Pelvic fracture can cause:

Massive internal bleeding

- **Mechanism of injury:**

- Road traffic accident
- Fall
- Crush injury

Pelvic fracture

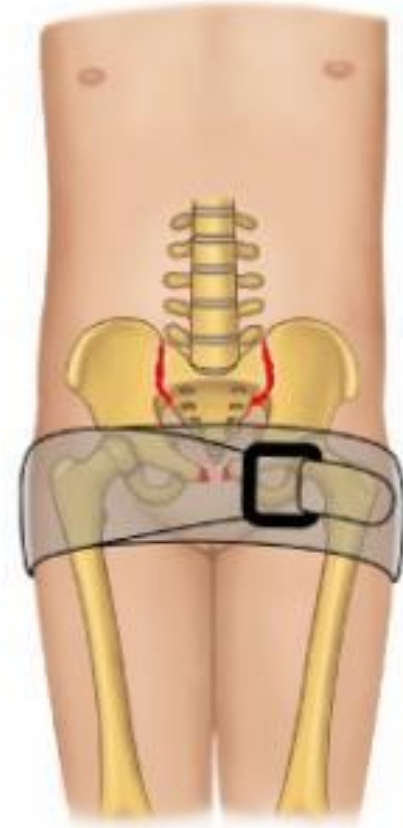
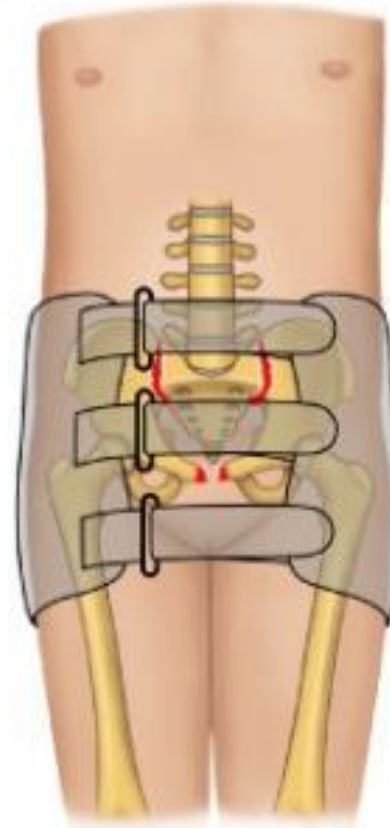
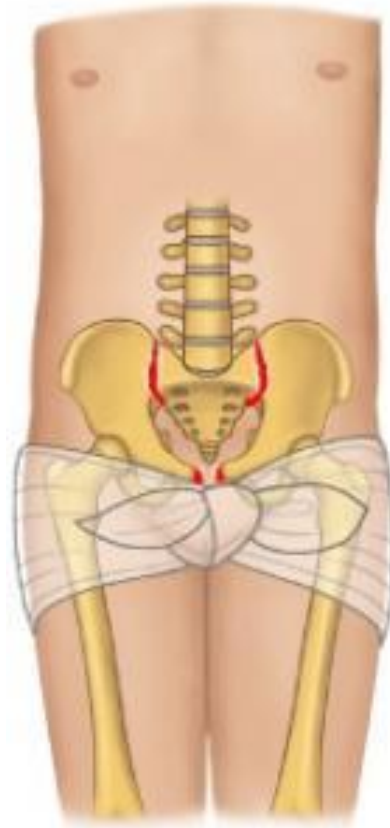
Signs of pelvic fracture:

- Pelvic pain
- Inability to stand
- Leg length difference
- Blood at urethral opening
- Hypotension (shock)



Pelvic fracture

- **Management of cervical fracture:**
- ABCDE
- Control bleeding if present
- Apply pelvic binder



Sprain

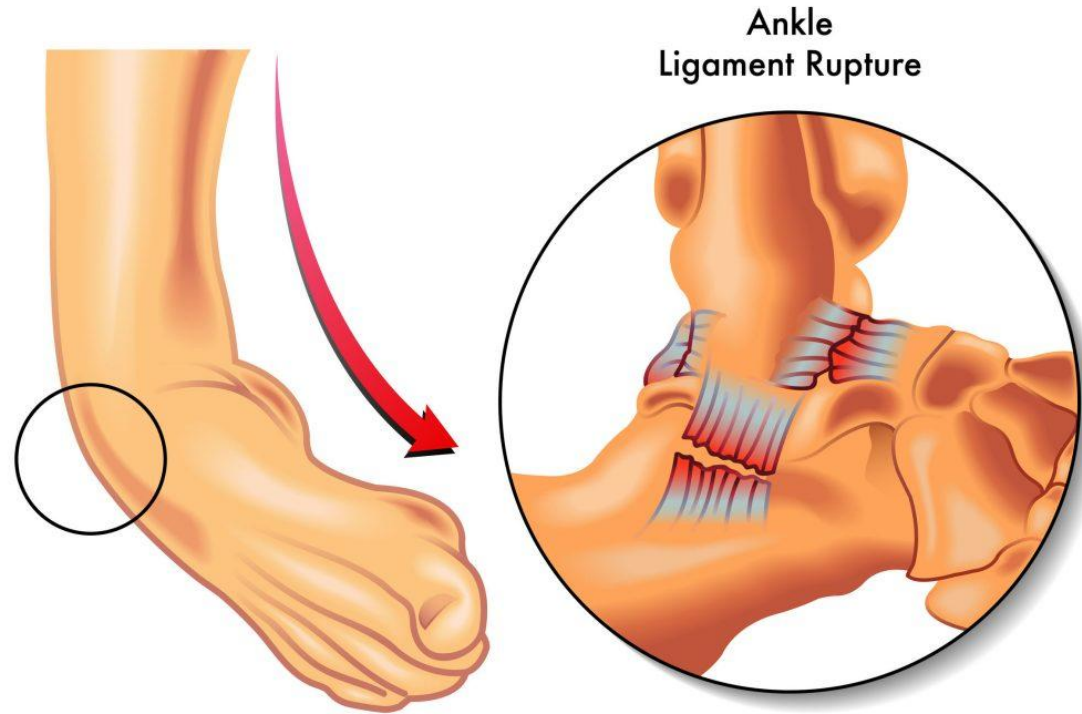
Ligament injury due to twisting or stretching

Common in:

Ankle

Knee

Wrist



Sprain

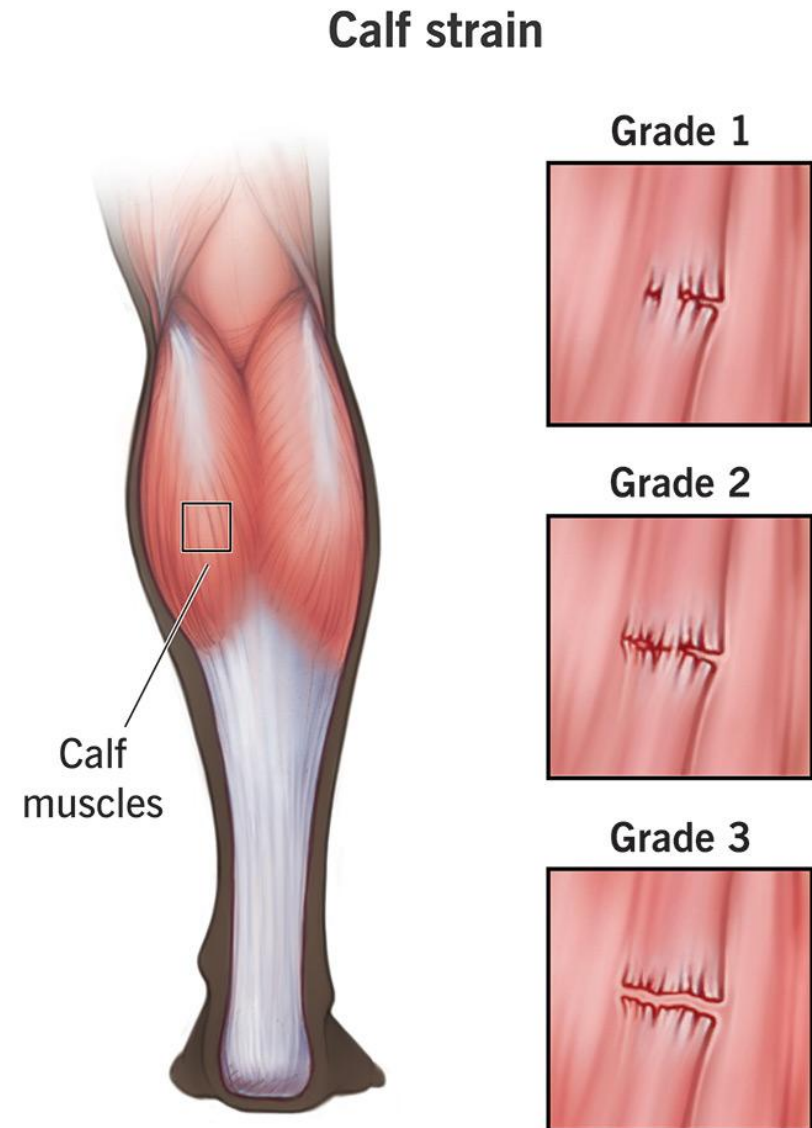
Presentation:

- Sudden twisting of the affected side
- Sudden intense pain
- Restricted mobility
- Swelling
- Bruising
- Tenderness over the affected site
- No bony tenderness



Strain

Muscle or tendon injury due to overstretching or tearing.



Strain

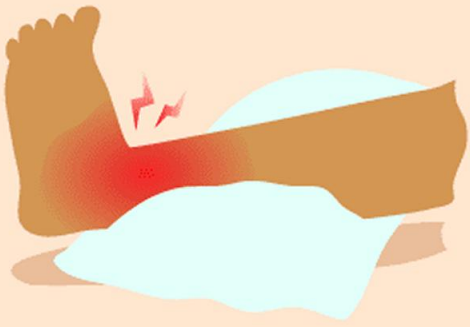
Presentation:

- Sudden sharp pain
- Lose of power in the affected area
- Tenderness over the affected area
- Swelling

Treatment

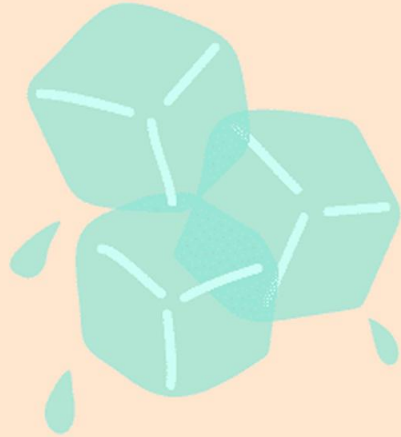
- **R:** Rest
- **I:** Ice
- **C:** Compression
- **E:** Elevation

R.I.C.E.



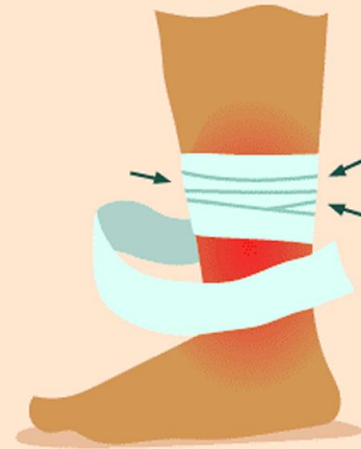
01 Rest

For the first
24 – 48 hours



02 Ice

For 20 min
4 – 8x per day



03 Compress

To reduce pain
& swelling



04 Elevate

6 – 10 in above
the heart

Feature	Fracture	Sprain	Strain
Pain	Severe	Moderate	Moderate
Deformity	Common	Rare	No
Bone tenderness	Yes	No	No
Ligament injury	No	Yes	No
Muscle injury	No	No	Yes
X-ray	Abnormal	Normal	Normal

Key takeaways

- ✓ Always assess ABCDE first
- ✓ Check neurovascular status
- ✓ Control bleeding
- ✓ Immobilize properly
- ✓ Check pulses before and after splint
- ✓ Refer red flags immediately

Red flags:

- ⚠ Neck pain after trauma
- ⚠ Loss of consciousness
- ⚠ Multiple injuries

Thank you